Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

THIRD SEMESTER B.TECH. (CIVIL ENGINEERING)

END SEMESTER MAKE-UP EXAMINATIONS, JANUARY 2025

ENGINEERING MATHEMATICS-III [MAT-2125]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 04 JANUARY 2025** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

Q.NO	Questions	Marks	CO	BTL
1A.	Represent the function $f(x) = x + x^2,\ -\pi \le x \le \pi$, $f(x + 2\pi) = f(x)$ as Fourier series.	4	1	3
1B.	Given $P(A) = \frac{1}{2}, P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{4}$. Evaluate $P\left(A/\bar{B}\right)$ and $P\left(\bar{A}/\bar{B}\right)$.	3	2	4
1C.	A two-dimensional random variable (X, Y) has the joint probability distribution function $f(x, y) = kxy, 0 \le x \le y, 0 \le y \le 4$ find 'k' and check whether X and Y are independent.	3	3	3
2A.	Check whether the given vector $\vec{F} = (2xy + z^3)i + x^2 j + 3xz^2 k$ is irrotational and find its scalar potential.	4	4	4
2B.	Obtain half-range cosine and sine series expansion of $f(x) = x, 0 \le x \le \pi$.	3	1	4
2C.	Find the angle between the surfaces $3x^2 - y^2 + 2z = 1$ and $z^2 + 3x = xy^2 z$ at the point $(1, -2, 1)$.	3	4	3
3A.	In a certain college, 25% of the boys and 10% of the girls wear spectacles. The girls constitute 60% of the student body. If a student is selected at random and is found to wear spectacles, determine the probability that the student is a girl?	4	2	3
3B.	Solve $u_{xx} - 2u_{xy} + u_{yy} = 0$ using the transformation $v = x, z = x + y$.	3	5	4

3C.	Find the value of n such that $\nabla.(r^n\vec{r}) = 0$, where r is a magnitude of position vector and n is a constant.	**3**	**4**	**4**
4A.	A 2-dimensional random variable (X, Y) has the joint pdf $f(x,y) = \begin{cases} 3xy(x+y), & 0 \le x \le 1, 0 \le y \le 1 \\ 0, & Otherwise \end{cases}$ compute the correlation coefficient ρ_{XY}.	**4**	**3**	**4**
4B.	Evaluate $\iint_S \vec{F}.\hat{n}\, ds$ where $\vec{F} = 4xzi - y^2j + yzk$ and S is the surface of the cube bounded by the planes $x = 0, x = 1, y = 0, y = 1, z = 0$ and $z = 1$.	**3**	**4**	**3**
4C.	Find the Fourier transform of $f(x) = 1, \lvert x\rvert < a$. Hence evaluate $\int_0^\infty \frac{sint}{t} dt$.	**3**	**1**	**4**
5A.	The probability distribution of a finite random variable X is given by the following table x_i: 0, 1, 2, 3, 4, 5, 6 $p(x_i)$: k, $3k$, $5k$, $7k$, $9k$, $11k$, $13k$ (a) Find k (b) Find $P\{3 < X \le 6\}$ (c) Find standard deviation σ	**4**	**3**	**4**
5B.	Solve $3\frac{\partial u}{\partial x} + 2\frac{\partial u}{\partial y} = 0$ by the method of separation of variables.	**3**	**5**	**3**
5C.	Compute the constant term and first harmonic of the Fourier series of $f(x)$ given in the following table. $x°$: 0, 45, 90, 135, 180, 225, 270, 315 y: 2, 1.5, 1, 0.5, 0, 0.5, 1, 1.5	**3**	**1**	**3**